

# DS&AI

## Machine Learning

### Regression

DPP: 1

- Q1** The relation between studying time (in hours) and grade on the final examination (0-100) in a random sample of students in the Introduction to Machine Learning Class was found to be :  
 $\text{Grade} = 30.5 + 15.2(h)$  How will a student's grade be affected if she studies for four hours ?
- (A) It will go down by 30.4 points.  
 (B) It will go down by 30.4 points.  
 (C) It will go up by 60.8 points.  
 (D) The grade will remain unchanged.  
 (E) It cannot be determined from the information given

- Q2** What does the slope coefficient in a linear regression model indicate?
- (A) The point where the regression line intersects the y-axis  
 (B) The dependent variable changes for every one-unit change in the independent variable  
 (C) The average value of the dependent variable  
 (D) The dispersion of the dependent variable

- Q3** Find the mean of squared error for the given predications :

| Y  | F(X) |
|----|------|
| 1  | 2    |
| 2  | 3    |
| 4  | 5    |
| 8  | 9    |
| 16 | 15   |
| 32 | 31   |

Hint : Find the squared error for each predication and take the mean of that.

- (A) 1 (B) 5

- (C) 1.5 (D) 5/6

- Q4** In the regression model ( $y = a + bx$ ) where  $x = 2.50$ ,  $y = 5.50$  and  $a = 1.50$  ( $x$  and  $y$  denote mean of variables  $x$  and  $y$  and  $a$  is a constant), which one of the following values of parameter 'b' of the model is correct?
- (A) 1.75 (B) 1.60  
 (C) 2.00 (D) 2.50

- Q5** What is the purpose of regularization in linear regression?
- (A) To make the model more complex  
 (B) To avoid underfitting  
 (C) To encourage overfitting  
 (D) To reduce the complexity of the model

- Q6** A set of observations of independent variable ( $x$ ) and the corresponding dependent variable ( $y$ ) is given below :

|   |    |    |    |    |
|---|----|----|----|----|
| X | 5  | 2  | 4  | 3  |
| Y | 16 | 10 | 13 | 12 |

Based on the data, the coefficient  $a$  of the linear regression model,  $y = a + bx$  is estimated as 6.1

The coefficient  $b$  is \_\_\_\_\_. (round off to one decimal place )

- Q7** The linear regression model  $y = a_0 + a_1x_1 + a_2x_2 + \dots + a_px_p$  is to be fitted to a set of  $N$  training data points having  $p$  attributes each. Let  $X$  be  $N \times (p + 1)$  vectors of input values (augmented by 1's),  $Y$  be  $N \times 1$  vector of target values, and  $\theta(0)$  be  $(p + 1) \times 1$  vector of parameter values ( $a_0, a_1, a_2, \dots, a_p$ ). If the sum squared error is minimized for obtaining the optimal regression model, which of the following equation holds ?



(A)  $X^T X = X Y$

(B)  $X A = X^T Y$

(C)  $X^T X A = Y$

(D)  $X^T X A = X^T Y$

- Q8** Use the regression equation to predict the glucose level given the age. Consider the following is the data set for understanding the concept of Linear Regression Numerical Example with One Independent Variable.

| SUBJECT | AGE-X | GLUCOSE LEVEL Y |
|---------|-------|-----------------|
| 1       | 43    | 99              |
| 2       | 21    | 65              |
| 3       | 25    | 79              |
| 4       | 42    | 75              |
| 5       | 57    | 87              |
| 6       | 59    | 81              |
| 7       | 55    | ?               |

- Q9** In linear regression, what is the primary difference between Lasso (L1 regularization) and Ridge (L2 regularization)?
- (A) Lasso tends to produce sparse coefficient vectors, while Ridge does not.
- (B) Ridge tends to produce sparse coefficient vectors, while Lasso does not.
- (C) Both Lasso and Ridge produce sparse coefficient vectors.
- (D) Both Lasso and Ridge tend to produce non-sparse coefficient vectors.

- Q10** Consider the following set of points:  $\{(-2, -1), (1, 1), (3, 2)\}$
- (a) Find the least square regression line for the given data points.
- (b) Plot the given points and the regression line in the same rectangular system of axes.

- Q11** In the table below, the  $x_i$  column shows scores on the aptitude test. Similarly, the  $y_i$  column shows statistics grades. The last two columns show deviations scores - the difference between the student's score and the average score on each measurement. The last two

rows show sums and mean scores.

Find the regression equation

| Student | $x_i$ | $y_i$ | $(x_i - \bar{x})^2$ | $(y_i - \bar{y})^2$ |
|---------|-------|-------|---------------------|---------------------|
| 1       | 95    | 85    | 289                 | 64                  |
| 2       | 85    | 95    | 49                  | 324                 |
| 3       | 80    | 70    | 4                   | 49                  |
| 4       | 70    | 65    | 64                  | 144                 |
| 5       | 60    | 70    | 324                 | 49                  |
| Sum     | 390   | 385   | 730                 | 630                 |
| Mean    | 78    | 77    |                     |                     |

- Q12** The values of  $y$  and their corresponding values of  $x$  are shown in the table below.

|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $x$ | 0 | 1 | 2 | 3 | 4 |
| $y$ | 2 | 3 | 5 | 4 | 6 |

- (a) Find the least square regression line  $y = a x + b$ .
- (b) Estimate the value of  $y$  when  $x = 10$ .

- Q13** When performing linear regression, multicollinearity can be problematic. Which of the following statements about multicollinearity is true?
- (A) Multicollinearity occurs when there is no correlation between independent variables.
- (B) Multicollinearity makes it easier to interpret the individual coefficients in the regression model.
- (C) Multicollinearity inflates the standard errors of the regression coefficients.
- (D) Multicollinearity always improves the predictive performance of the model.

- Q14** Suppose you have  $m = 14$  training examples with  $n = 3$  features (excluding the additional all-ones feature for the intercept term, which you should add), the normal equation is  $0 = (X^T X)^{-1} X^T Y$ . For the given values of  $m$  and  $n$ , what are the dimensions of  $\theta$ ,  $X$  and  $y$  in this equation?



- (A)  $X$  is  $14 \times 4$ ,  $y$  is  $14 \times 1$ ,  $q$  is  $4 \times 1$   
 (B)  $X$  is  $14 \times 3$ ,  $y$  is  $14 \times 1$ ,  $q$  is  $3 \times 3$   
 (C)  $X$  is  $14 \times 4$ ,  $y$  is  $14 \times 4$ ,  $q$  is  $4 \times 4$   
 (D)  $X$  is  $14 \times 3$ ,  $y$  is  $14 \times 1$ ,  $q$  is  $3 \times 1$

- Q15** The closed form solution for linear regression is  $0 = (X^T X)^{-1} X^T y$ . Suppose you have  $N = 35$  training examples and  $M = 5$  features (excluding the bias term). Once the bias term is now included, what are the dimensions of  $X$ ,  $y$ ,  $q$  in the closed form equation?  
 (A)  $X$  is  $35 \times 6$ ,  $y$  is  $35 \times 1$ ,  $q$  is  $6 \times 1$   
 (B)  $X$  is  $35 \times 6$ ,  $y$  is  $35 \times 6$ ,  $q$  is  $6 \times 1$   
 (C)  $X$  is  $35 \times 5$ ,  $y$  is  $35 \times 1$ ,  $q$  is  $5 \times 1$   
 (D)  $X$  is  $35 \times 5$ ,  $y$  is  $35 \times 5$ ,  $q$  is  $5 \times 5$

- Q16** Suppose the score of the midterm, homework, and final of 4 students are given by

| Midterm | Homework | Final |
|---------|----------|-------|
| 80      | 85       | 96    |
| 72      | 68       | 74    |
| 94      | 97       | 90    |
| 69      | 70       | 78    |

You'd like to use linear regression to predict a student's final exam score from his/her midterm exam score and homework score. The normal equation of linear regression is  $q = (X^T X)^{-1} X^T y$ . What are  $X$  and  $y$  in this problem.

- Q17** Assume that you have a dataset composed of  $N$  observations: the target  $t$  and features  $X$ . You want to fit a linear regression model and find the weights  $w$ , but you also know that more data is always helpful. Instead of fitting a model with  $t \in \mathbb{R}^n$  and  $X \in \mathbb{R}^{n \times d}$ , you concatenate the data fit a model using

$$\begin{bmatrix} t \\ t \end{bmatrix} \in \mathbb{R}^{2n} \text{ and } \begin{bmatrix} X \\ X \end{bmatrix} \in \mathbb{R}^{2n \times d}.$$

Running linear regression on this new dataset will give the same weights as on the original dataset.

- Q18** First, we use a linear regression method to model this data. To test our linear regressor, we choose at random some data records to be a training set, and choose at random some of the remaining records to be a test set. Now let us increase the training set size gradually. As the training set size increases, what do you expect will happen with the mean training and mean testing errors?  
 (A) Mean Training Error: Increase  
 Mean Testing Error: Decrease  
 (B) Mean Training Error: Increase  
 Mean Testing Error: Increase  
 (C) Mean Training Error: Decrease  
 Mean Testing Error: Increase  
 (D) Mean Training Error: Decrease  
 Mean Testing Error: Decrease

- Q19** Suppose now that Bob records each feature in a different unit; that is, for each coordinate  $z_j^{(i)} = c_j x_j^{(i)}$  for all  $i$ , where  $c_j > 0$  is a scalar, and the  $c_j$ 's are not all equal. Do Alice and Bob still have the same training loss and is  $w_{\text{Alice}} = w_{\text{Bob}}$ ? If yes, justify your answer. If no, provide an example of a dataset where this is not the case.



# Answer Key

Q1 C  
Q2 B  
Q3 D  
Q4 B  
Q5 D  
Q6 1.9~1.9  
Q7 D  
Q8 86.327~86.327  
Q9 A  
Q10 0.2631~0.2631

Q11 26.78~26.78  
Q12 11.2~11.2  
Q13 C  
Q14 A  
Q15 A  
Q16 00  
Q17 00  
Q18 A  
Q19 0



# Hints & Solutions

Note: scan the QR code to watch video solution

## Q1 Text Solution:

$$G = 30.5 + 15.2W$$

Student 4 hour

$$G \text{ will be effected as } 15.5 \times 4 = 60.8$$

## Q2 Text Solution:

$$y = mx + c \text{ (1D data)}$$

Shop

$$m = \frac{dy}{dx} \Rightarrow dy = m dx \Rightarrow \text{If } dx = 1 \text{ i.e.}$$

change in indepndent variable = 1 then dy = m

## Q3 Text Solution:

| error y-y' | error <sup>2</sup> |
|------------|--------------------|
| -1         | 1                  |
| -1         | 1                  |
| -1         | 1                  |
| -1         | 1                  |
| 1          | 1                  |
| 0          | 0                  |

y: actual value

f(x): predicted value

mean of error<sup>2</sup>

$$\Rightarrow \frac{1+1+1+1+1+0}{6} \Rightarrow \frac{5}{6}$$

## Q4 Text Solution:

We know that

$$y = a + bx$$

$$\text{So, } \bar{y} = a + b\bar{x}$$

$$b = \frac{\bar{y} - a}{\bar{x}} \Rightarrow \frac{5.5 - 1.5}{2.5} = 1.6$$

## Q5 Text Solution:

Regularisation

1. Make model less complex
2. oviad ovefitting
3. Reduces complexity of model

## Q6 Text Solution:

$$a = 6.1,$$

We know that

$$\bar{y} = a + b\bar{x}$$

$$b = \frac{\bar{y} - a}{\bar{x}}$$

$$b = 1.9$$

$$\bar{x} = 7/2$$

$$\bar{y} = \frac{51}{4}$$

## Q7 Text Solution:

done is class

$$(X^T X A = X^T Y)$$

## Q8 Text Solution:

So, hre we have to geate model

$$y = mx + c$$

$$x = \begin{bmatrix} 1 & 43 \\ 1 & 21 \\ 1 & 25 \\ 1 & 42 \\ 1 & 57 \\ 1 & 59 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 6 & 247 \\ 247 & 11409 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 486 \\ 20485 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 11409 & -247 \\ -247 & 6 \end{bmatrix} \frac{1}{7445}$$

$$\begin{bmatrix} c \\ m \end{bmatrix} = (X^T X)^{-1} [X^T Y] =$$

$$\begin{bmatrix} 11409 & -247 \\ -247 & 6 \end{bmatrix} \begin{bmatrix} 486 \\ 20485 \end{bmatrix} \frac{1}{7445}$$

$$\begin{bmatrix} 484979 \\ 2868 \end{bmatrix} \frac{1}{7445}$$

$$\Rightarrow C = 65.14$$



$$m = .3852$$

$$\text{So, @ } x = 55, y = mx + c \Rightarrow 86.32$$

### Q9 Text Solution:

Lasso Sparse sol

Because it send to make more coefficients B's to zero

### Q10 Text Solution:

$$\left. \begin{array}{l} x - 2 \quad 1 \quad 3 \\ y - 1 \quad 1 \quad 2 \end{array} \right\} \text{Model } y = mx + c$$

So,

$$x = \begin{bmatrix} 1 & -2 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} X^T X = \begin{bmatrix} 3 & 2 \\ 2 & 14 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 14 & -2 \\ -2 & 3 \end{bmatrix} \frac{1}{38}$$

$$X^T X = \begin{bmatrix} 2 \\ 9 \end{bmatrix}, \begin{bmatrix} c \\ m \end{bmatrix} = (X^T X)^{-1} (X^T Y) = \begin{bmatrix} 10/38 \\ 23/38 \end{bmatrix}$$

$$C = \frac{10}{38}$$

$$M = \frac{23}{38}$$

$$y = mx + c$$

$$y = .605x + .2631$$

### Q11 Text Solution:

$$y = mx + c$$

$$m = \frac{\text{cov}(x, y)}{\sigma_x^2}$$

$$\text{cov}(x, y) = \sum_{i=1}^5 (x_i - \bar{x})(y_i - \bar{y})$$

$$\Rightarrow 17 \times 8 + 7 \times 18 + 2 \times (-2) - 8 \times -12 + -18 \times \frac{7}{5}$$

$$\text{cov} = \frac{470}{5} = 94$$

$$m = \frac{\text{cov of } x, y}{\text{var of } x} = \frac{94}{\frac{\sum (x_i - \bar{x})^2}{5}} = \frac{94}{\frac{730}{5}} = .6438$$

$$c = \bar{y} - m\bar{x} \Rightarrow 77 - .6438 \times 78 \Rightarrow 26.78$$

$$\text{So, } (y = .6438x + 26.78)$$

### Q12 Text Solution:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \Rightarrow X^T X = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{50} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 20 \\ 49 \end{bmatrix}$$

$y = ax + b$  is model

$$\begin{bmatrix} b \\ a \end{bmatrix} = (X^T X)^{-1} (X^T Y) = \begin{bmatrix} 110 \\ 45 \end{bmatrix} \frac{1}{50}$$

$$b = \frac{110}{50}$$

$$a = \frac{45}{50}$$

$$y = .9x + 2.2$$

$$\text{@ } x = 10$$

$$y = 9 + 2.2 = 11.2$$

### Q13 Text Solution:

Due to multicollincantly model bcome poor.

### Q14 Text Solution:

$$X = 14 \text{ rows } 14 \times 4$$

$$4 \text{ column}$$

$$X = 14 \text{ Rows } 14 \times 1$$

$$1 \text{ column}$$

$$Q = Q_0 \quad Q_3$$

$$4 \text{ row } 4 \times 1$$

$$1 \text{ column}$$

### Q15 Text Solution:

we have to predict  $y$  final core using mid

terms & homework soure 2 determines  $x^1$ ,  $x^2$ .

Final score  $y$ , mid term home work are  $x$



**Q16 Text Solution:**

data

$$\begin{matrix} P_1 & \begin{bmatrix} X_1 & Y_1 \\ X_2 & Y_2 \end{bmatrix} \\ P_2 & \end{matrix}$$

$$\text{loss } f_{\text{xn}} \sum_{i=1}^2 (y_i - y_i^1)^2$$

Now new data

data

$$P_1 \quad x_1 \quad y_1$$

$$P_2 \quad x_2 \quad y_2$$

$$P_3 \quad x_1 \quad y_1$$

$$P_4 \quad x_2 \quad y_2$$

$$\text{Now, } \alpha = \sum_{i=1}^4 (y_i - y_i)^2$$

$$= 2 \sum_{i=1}^2 (y_i - y_i)^2$$

Since loss fcn in both case is same (only multiplied by 2 in case 2) the result to minimize L will be same

yes the model will be same.

**Q17 Text Solution:**

1000points

Test data

Training data

Now, if we increase the training data what will happen

- our target is to find pattern of data, if we have large data then we get better fit.
- When training data is very small there mode over fit.

|                          |                           |
|--------------------------|---------------------------|
| Training data very small | Training data increase    |
| • Over fit model         | • Model generative better |
| • Training error = 0     | • Training error $\neq 0$ |
| • Testing error is high  | • Testing error reduce    |

**Q18 Text Solution:**

1D data

$$\begin{matrix} X & Y \\ x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{matrix}$$

Model  
 $y = mx + c$ 

Modify data

1D data

$$\begin{matrix} X & Y \\ c_1 x_1 & y_1 \\ c_2 x_2 & y_2 \\ \vdots & \vdots \\ c_1 x_N & y_N \end{matrix}$$

Since y value has not changed but x has changed, So for same y we need to update the coeff 'm' to get same y for new x

$$W_{\text{Alice}} \neq W_{\text{Bob.}}$$

**Q19 Text Solution:**

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